Name _____ Period _____ Date _____

2-6 Lesson Quiz: The Quadratic Formula

short form

1. Solve $x^2+8x-6=0$ using the Quadratic Formula.

Ⓐ $x = 32 + 2\sqrt{2}$ and $x = 32 - 2\sqrt{2}$

Ⓑ $x = -4 + \sqrt{22}$ and $x = -4 - \sqrt{22}$

Ⓒ $x = 12$ and $x = -\frac{1}{2}$

Ⓓ $x = 4 + 2\sqrt{22}$ and $x = 4 - 2\sqrt{22}$

Show your work: $a =$ ____, $b =$ ____, $c =$ ____

2. Solve $x^2+2x+7=0$ using the Quadratic Formula. Select **all** solutions that apply.

☐ **A.** $x = -1 + i\sqrt{6}$

☐ **B.** $x = -1 + \sqrt{6}$

☐ **C.** $x = -1 - i\sqrt{6}$

☐ **D.** $x = -1 - \sqrt{6}$

☐ **E.** $x = 1 - \sqrt{6}$

Show your work:

5. What value(s) of b will cause $27x^2 + bx + 3 = 0$ to have one real solution? Select **all** that apply.

☐ **A.** $b = -18$

☐ **B.** $b = -9$

☐ **C.** $b = -3$

☐ **D.** $b = 3$

☐ **E.** $b = 9$

☐ **F.** $b = 18$

Show your work: $a =$ ____, $c =$ ____

$b^2 - 4ac =$ ____ (one real solution)

Explain your choice in one sentence using the word *discriminant*:

Scoring: Question 1 and Question 5 are 10 points each. Question 2 is a bonus of up to 5 points (both correct boxes, none wrong). Total out of 20.

2-6 Lesson Quiz, short form: answer key

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1. **B,** $x = -4 \pm \sqrt{22}$**.** $a = 1$, $b = 8$, $c = -6$: $x = \dfrac{-8 \pm \sqrt{64 + 24}}{2} = \dfrac{-8 \pm \sqrt{88}}{2} = \dfrac{-8 \pm 2\sqrt{22}}{2} = -4 \pm \sqrt{22}$. Savvas quiz item 1, DOK 2, Realize skills A31, A72; IXL YQH (Example 1A). **10 points:** 6 for B, 4 for a, b, c and a correct substitution.

2. **A and C,** $x = -1 \pm i\sqrt{6}$**.** $a = 1$, $b = 2$, $c = 7$: $x = \dfrac{-2 \pm \sqrt{4 - 28}}{2} = \dfrac{-2 \pm \sqrt{-24}}{2} = \dfrac{-2 \pm 2i\sqrt{6}}{2} = -1 \pm i\sqrt{6}$. B and D drop the i (the radicand -24 is negative); E has the wrong sign on -1 and also drops the i. Savvas quiz item 2, DOK 2, Realize skills A31, A72; IXL YQH (Example 1B). **Bonus 5:** A and C and no other box; 3 for one of A or C and no wrong box; 0 otherwise.

5. **A and F,** $b = \pm 18$**.** $a = 27$, $c = 3$. One real solution when the discriminant is 0: $b^2 - 4(27)(3) = 0$, $b^2 - 324 = 0$, $b^2 = 324$, $b = \pm 18$. B and E (± 9) and C and D (± 3) give $81 - 324 < 0$ and $9 - 324 < 0$: two non-real solutions. Savvas quiz item 5, DOK 2, Realize skill A32; IXL QHK (Example 5). **10 points:** 6 for A and F and no other box (3 for one of them and no wrong box), 2 for $b^2 - 4(27)(3) = 0$, 2 for the sentence naming the discriminant.

Items left off this form (available for a retake or Form B): Savvas item 3, the nature of the roots of $49x^2 - 28x + 4 = 0$ (answer B, one real root: $(-28)^2 - 4(49)(4) = 784 - 784 = 0$; DOK 1, A32; taught by Example 3, cut; the packet's Key idea box and Example 5 use the same rule); Savvas item 4, Richard's ball, $h(t) = -5t^2 + 10t + 6$: does it reach 12 m? (answer D, the ball does not reach 12 m: $-5t^2 + 10t - 6 = 0$ has discriminant $100 - 120 = -20 < 0$; DOK 2, A31, A72; taught by Example 4, cut, and rehearsed by the packet's Practice page).

IXL replacement: SmartScore 70 or above on YQH and QHK replaces this quiz with 20/20.